APPLIED MULTIVARIATE METHODS833:525 / 970:527
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Homework 03 (50 points)

*HW must be **typed** in an appropriate word processing software application.*
Responses should be numbered and in chronological order.
Indicate your full name on the top of the document.
***Save a PDF of the completed document** with your first initial and last name, i.e., HW03_swilliams.pdf*
***Upload** the HW via Canvas before the due date.*
***Show only key steps of your calculation.** You get partial credit for showing work in case of wrong answer.*
Any hypothesis test must include a null and alternative hypothesis, test statistics, degrees of freedom (if applicable),decision and interpretation.

Problem 1:
Dataset to be used: `esoph_cancer.csv`
Please do not hand in computer output separately from responses to the questions below. Instead, cut and paste relevant output. Copy and paste (only relevant) R codes at the end of your homework in an appendix (annotated and in chronological order).

a) We would like to include the categorical variable, *tobgrp*, in the analysis to operationalize tobaccouse. Please explain *why* dummy variables would be useful. (4 points)

The original TOBACCO variable for tobacco use is measured at continuous ratio level of measurement the model with continuous variable will only explain how alcohol consumption will change with increase or decrease in Tobacco use controlling for age (see summary result model2). The summary shows consumption will increase using more tobacco. However, operationalize tobacco use in categorical ordinal level of measurement with generating dummy variables will give us better understanding of how increase in tobacco use will disproportionately affect the consumption of Alcohol, based on amount of different level of tobacco intake. It can also tell us better comparison with the baseline i.e. no tobacco use. Further, dummy variable can tell us better in decision making comparing it with different levels of slopes for each level of tobacco use.

b) Now describe how you would create the dummy variables. (6 points)

To generate dummy variables while adhering to the k-1 rule, where k is the number of dummy variables one wants to generate. Cleaning the TOBACCO variable by assigning NA to records where TOBACCO was 8 or 9. Then, I categorized the TOBACCO variable into four groups: non-use (0), low use (1), moderate use (2), and high usage (3), based on the quantity of tobacco consumed. With the rule it takes 4 -1=3 dummy variables where no use with four distinct levels of tobacco use, including the non-use category, it was unnecessary to create a separate dummy variable for the baseline group in the regression model. Instead, I created three dummy variables: tobgrp1, tobgrp2, and tobgrp3. These variables correspond to the three active levels of tobacco use—low, moderate, and high.

To achieve this, I utilized the ifelse() function in R to assign values conditionally. For each record in the dataset, the function checks the tbg value and assigns a 1 to the corresponding dummy variable if the record's tobacco use matches the group. For instance, if tobgrp equals 1, indicating low usage, tobgrp1 is set to 1, while tobgrp2 and tobgrp3 are set to 0. Similarly, for a tobgrp value of 2, tobgrp2 is set to 1, and the others to 0; and for a tobgrp value of 3, tobgrp3 is set to 1, with the rest set to 0. If none of the conditions for low, moderate, or high use are met (i.e., the tobgrp value is 0), all three dummy variables will be set to 0, indicating that the record falls into the baseline non-use category. See Annexure 1

c) Write down the population model you plan to estimate. (4 points)

Alcohol Consumption = $\beta 0 + \beta 1(\text{Age}) + \beta 2\ (\text{tbg1}) + \beta 3(\text{tbg2}) + \beta 4(\text{tbg3})$
$\beta 0$ represents the intercept of the model, whereas β1 shows slope for with age given no tobacco use, $\beta 2$ shows slope for low use (1 or 2) (tbg1) + $\beta 3$ shows slope for moderate (3 or 4) (tbg2) + $\beta 4$ shows slope for high (5,6 or7) (tbg3)

d) Test whether the model for estimating alcohol consumption based on tobacco use adjusted by age is meaningful on level of significance 5%. State the appropriate null and alternative hypothesis, test statistics and associated dfs,p-value, conclusion, and interpretation. Include the appropriate output in the answer. (7 points)

The model for estimating alcohol consumption based on tobacco use adjusted by age ismeaningful, we'll consider significance of the model using an F-test omnibus test.

Null Hypothesis (H_0):
The findings indicate that the model containing predictors AGE, tbg1, tbg2, and tbg3 does not improve the fit compared to a model with only the intercept. This suggests that neither AGE nor the tobacco use categories (tbg1, tbg2, tbg3) exhibit a relationship with alcohol consumption. Consequently, all slopes in the model are equivalent to zero, indicating no significant association between these variables and alcohol consumption.
$$H0 = \beta 1 = \beta 2 = \beta 3 = \beta 4 = 0$$

Alternative Hypothesis (H_1):
The observation suggests that the model including predictors AGE, tobgrp1, tobgrp2, and tobgrp3 yields a superior fit compared to the model with only the intercept. This implies that there is evidence of a association between at least one of the predictors (AGE, tobgrp1, tobgrp2, or tobgrp3) and alcohol consumption.
H1 = At least one $\beta j \neq 0$ for $j = 1,2,3,4$

The F-statistic and associated degrees of freedom (dfs) are used to determine the p-value, which can then becompared to the significance level to draw a conclusion.

F-statistic: 8.109, Degrees of Freedom: Model df = 4 (number of predictors), Residual df = 966 (number of observations -number of parameters), p-value: 1.971e-06. Since the p-value (1.971e-06) is much less than 0.05, we reject the null hypothesis. It means model is meaningful. **Conclusion and Interpretation:** There is significant evidence at the 5% level to conclude that the model for estimating alcoholconsumption based on tobacco use, adjusted by age, is meaningful. The outcome suggests that, as per the data and the model, there exists a statistically significant correlation between tobacco use categories and alcohol consumption when adjusting for age. However, it's crucial to recognize that although the model demonstrates statistical significance, the Multiple R-squared value of 0.03249 indicates that merely about 3.25% of the variance in alcohol consumption is elucidated by the model. This implies that there are other unaccounted factors contributing to the variability in alcohol consumption beyond what is included in the model. The adjusted r-squared is only 2.848% which means with adding additional variables it reduces from 3.2 to 2.8 %.

e) Test whether tobacco use is a significant predictor of alcohol consumption, while controlling for age. State the appropriate null and alternative hypotheses, test statistic and associated dfs, p-value, conclusion, and interpretation. Include the appropriate output in the answer. (7 points)

I have tested the overall impact of tobacco use on alcohol consumption while controlling for age.In order to this in a simpler way, I did the following to get my R output.
I tested the original variable TOBACCO with any use of tobacco or not in Model2 output:

A linear regression model using the lm function. The model predicts alcohol consumption (ALCOHOL) as a function of age (AGE) and the tobacco use variable (TOBACCO).

Now answering the question:

Hypothesis Testing
Null Hypothesis (H_0): $\beta TOBACCO=0$.
This states that there is no effect of tobacco use (any use vs. non-use) on alcohol consumption, after controllingfor age.
Alternative Hypothesis (H_1): $\beta TOBACCO \neq 0$.
This suggests that tobacco use does have an effect on alcohol consumption, differentiating between any use andnon-use, with age accounted for.

Test Statistic

- **Coefficient Tobacco** : $\beta TOBACCO=4.248$
- **Standard Error** : 0.7993
- **t-value**: 5.30
- **p-value**: $1.36\times10-7$
- **Degrees of Freedom** : 975 (based on the number of observations minus the number of parameters

Given the p-value of $1.36\times10-7.02\times10-6$ for the significance level coefficient, which is significantly less than the of 0.05, we reject the null hypothesis $H0$. **Tobacco**

The rejection of H0 indicates that there is a statistically significant effect of tobacco use on alcohol consumptionwhen controlling for age.

f) Which tobacco use groups are significantly different from non-smokers, while controlling for age, on a 5% level of significance? State both null and alternative hypotheses, test statistics and associated dfs, p-values, conclusions, and interpretations. Include the appropriate output in the answer. (15 points)

To determine which tobacco, use groups are significantly different from non-smokers (the reference category) while controlling for age, you can examine the significance levels of the coefficients for tbg1, tbg2, and tbg3 in the regression output. If the p-value associated with the coefficient for a specific tobacco use group is less than the chosen significance level (e.g., $\alpha = 0.05$), then we can infer that the corresponding group is significantly different from non-smokers. For instance, if the p-value for tbg1 is less than 0.05, it suggests that the group represented by tbg1 is significantly different from non-smokers. Similarly, if the p-value for tbg2 or tbg3 is less than 0.05, it indicates that the respective group is significantly different from non-smokers. Therefore, by examining the significance levels of the coefficients for tbg1, tbg2, and tbg3, you can determine which tobacco use groups exhibit a significant difference in alcohol consumption compared to non-smokers while controlling for age.
Hypotheses for each tobacco use group

For tbg1 (low tobacco use):
$H0: \beta tbg1 = 0$ (Low tobacco use is not significantly different from non-smokers in terms of alcohol consumption)
$H1: \beta tbg1 \neq \neq \neq 0$(low tobacco use is significantly different from non-smokers in terms of alcohol consumption)

For tbg2 (moderate tobacco use):

H0:β tbg2=0 (moderate tobacco use is not significantly different from non-smokers)
H1 :β tbg2≠0 (moderate tobacco use is significantly different from non-smokers)

For tbg3 (high tobacco use):

$H0:\beta tbg3=0$ (high tobacco use is not significantly different from non-smokers)(high tobacco use is significantly different from non-smokers)
$H1:\beta tbg3\neq 0$

Degrees of Freedom (df): 966
tbg1: Slope**:** 11.4057 **Std. Error:** 4.7502 **t-value**: 2.401 **p-value:** 0.0165
tbg2: **Slope:** 19.7705 **Std. Error:** 4.8259 **t-value:** 4.097 **p-value:** 4.54e-05
tbg3: **Slope:** 24.8921 **Std. Error:** 4.9561 **t-value:** 5.022 **p-value:** 6.07e-07

tbg1(low tobacco use):

With a p-value of 0.0165, which is less than the chosen significance level of 0.05, we reject the null hypothesis (H0) for tbg1. This indicates that low tobacco use (1-9 grams/day) is significantly different from non-smokers in terms of alcohol consumption when age is controlled for. In other words, individuals who smoke 1-9 grams/day consume significantly more alcohol than non-smokers, adjusting for age.
tbg2 (moderate tobacco use):

Indeed. With a p-value of 4.54e-05 (which is much lower than 0.05), we reject the null hypothesis (H0) for tbg2. This indicates that moderate tobacco use (10-19 grams/day) is significantly different from non-smokers in terms of alcohol consumption, with age taken into account. In other words, individuals who smoke 10-19 grams/day consume significantly more alcohol than non-smokers, even after adjusting for age.

tbg3 (high tobacco use):

The p-value (6.07e-07) leads us to reject H0 for tbg3, indicating a significant difference from non-smokers. Individuals in the high tobacco use category (20+ grams/day) have significantly greater alcoholconsumption than non-smokers, when controlling for age. These results demonstrate that all three groups of tobacco users—low, moderate, and high—consume significantly more alcohol than non-smokers, with statistical adjustments made for age. This highlights theassociation between tobacco use and increased alcohol consumption within the study population.

g) Test whether age is a significant predictor of alcohol consumption when adjusted by tobacco use at the 5% level of significance. State the appropriate null and alternative hypothesis, test statistics and associated dfs, p-value, conclusion, and interpretation. Include the appropriate output in the answer.(7 points)

Hypotheses

Null Hypothesis (H_o): βAGE=0. Age does not have a significant effect on alcohol consumption when tobaccouse is controlled for.

Alternative Hypothesis (H_1): βAGE≠0. Age has a significant effect on alcohol consumption when tobacco use is controlled for.

Test Statistic Estimate for AGE: Slope :0.1438
Standard Error for AGE: 0.1253

T-value for AGE: 1.148

P-value for AGE: 0.2512

Degrees of Freedom: 966 (based on the number of observations minus the number of estimated parameters)

Given the p-value of 0.2512 or 25.12 % for the AGE coefficient, which is greater than the 5% level of significance (0.05), wefail to reject the null hypothesis *H*0.

The failure to reject the null hypothesis (H0) implies that, within the context of this model, age is not a significant predictor of alcohol consumption when tobacco use is taken into account. This suggests that, after adjusting for the levels of tobacco use represented by tobgrp1, tobgrp2, and tobgrp3, the impact of age on alcohol consumption is not statistically significant at the 5% significance level. In other words, once tobacco use is considered, age does not appear to have a significant effect on alcohol consumption in this model.

Appendix

```
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate")
library(readr)
c.cases <- read_csv("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate/esoph_cancer.csv")
esoph_cancer <- read_csv("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate/esoph_cancer.csv")
esoph_cancer$tobgrp <- ifelse(esoph_cancer$TOBACCO == 8 | esoph_cancer$TOBACCO == 9, NA,

                              ifelse(esoph_cancer$TOBACCO == 0, 0,

                                     ifelse(esoph_cancer$TOBACCO >= 5, 3,

                                            ifelse(esoph_cancer$TOBACCO >= 3, 2,

                                                   ifelse(esoph_cancer$TOBACCO >= 1, 1, NA)))))

esoph_cancer$tbg1 <- ifelse(esoph_cancer$tobgrp == 1,1,0)
esoph_cancer$tbg2 <- ifelse(esoph_cancer$tobgrp == 2,1,0)
esoph_cancer$tbg3 <- ifelse(esoph_cancer$tobgrp == 3,1,0)
model1 <- lm(ALCOHOL ~ AGE + tbg1 + tbg2 + tbg3, data = esoph_cancer)
summary (model1)
model2 <- lm (ALCOHOL ~ AGE + TOBACCO, data = esoph_cancer)
summary (model2)
```

```
Call:
lm(formula = ALCOHOL ~ AGE + tbg1 + tbg2 + tbg3, data = esoph_cancer)

Residuals:
   Min     1Q Median     3Q    Max
-67.78 -28.10  -7.55  18.31 851.68

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.3916     6.8989   4.840 1.51e-06 ***
AGE           0.1438     0.1253   1.148   0.2512
tbg1         11.4057     4.7502   2.401   0.0165 *
tbg2         19.7705     4.8259   4.097 4.54e-05 ***
tbg3         24.8921     4.9561   5.022 6.07e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 53.41 on 966 degrees of freedom
  (7 observations deleted due to missingness)
Multiple R-squared:  0.03249,   Adjusted R-squared:  0.02848
F-statistic: 8.109 on 4 and 966 DF,  p-value: 1.971e-06
```

```
Call:
lm(formula = ALCOHOL ~ AGE + TOBACCO, data = esoph_cancer)

Residuals:
   Min     1Q Median     3Q    Max
-86.33 -28.41  -8.08  18.48 853.20

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.9300     6.8216   4.974 7.75e-07 ***
AGE           0.1822     0.1218   1.496    0.135
TOBACCO       4.2437     0.7993   5.309 1.36e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 53.32 on 975 degrees of freedom
Multiple R-squared:  0.03071,   Adjusted R-squared:  0.02873
F-statistic: 15.45 on 2 and 975 DF,  p-value: 2.485e-07
```